ASSESSMENT -4

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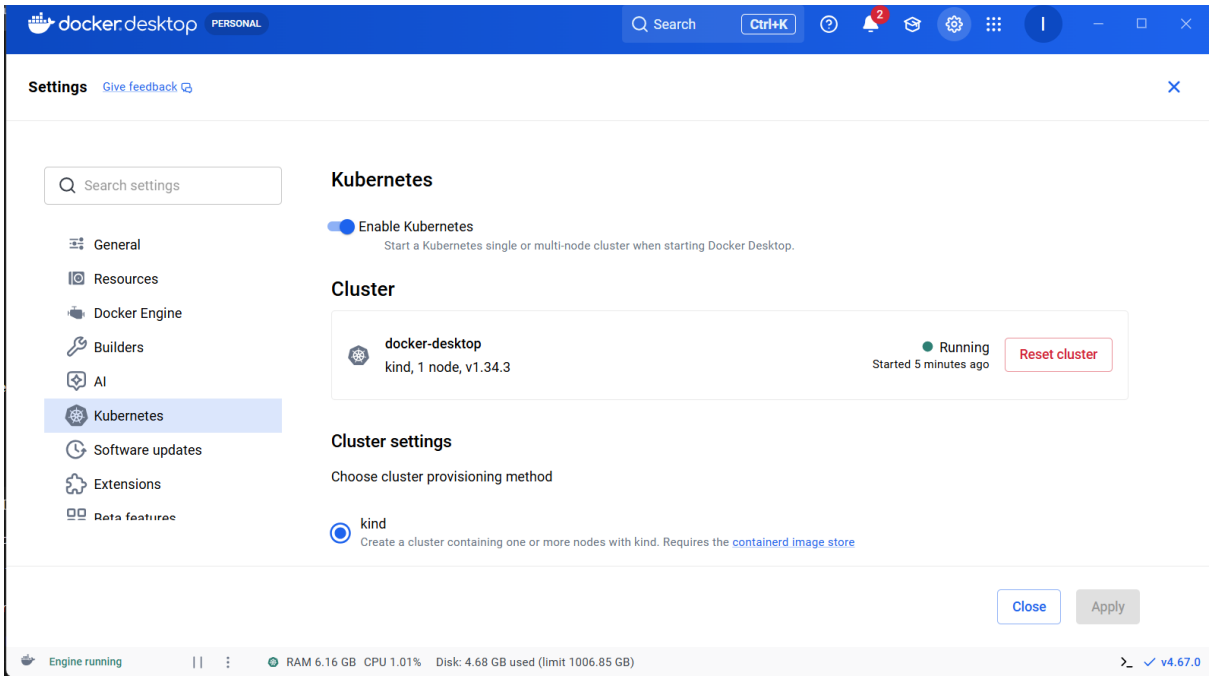
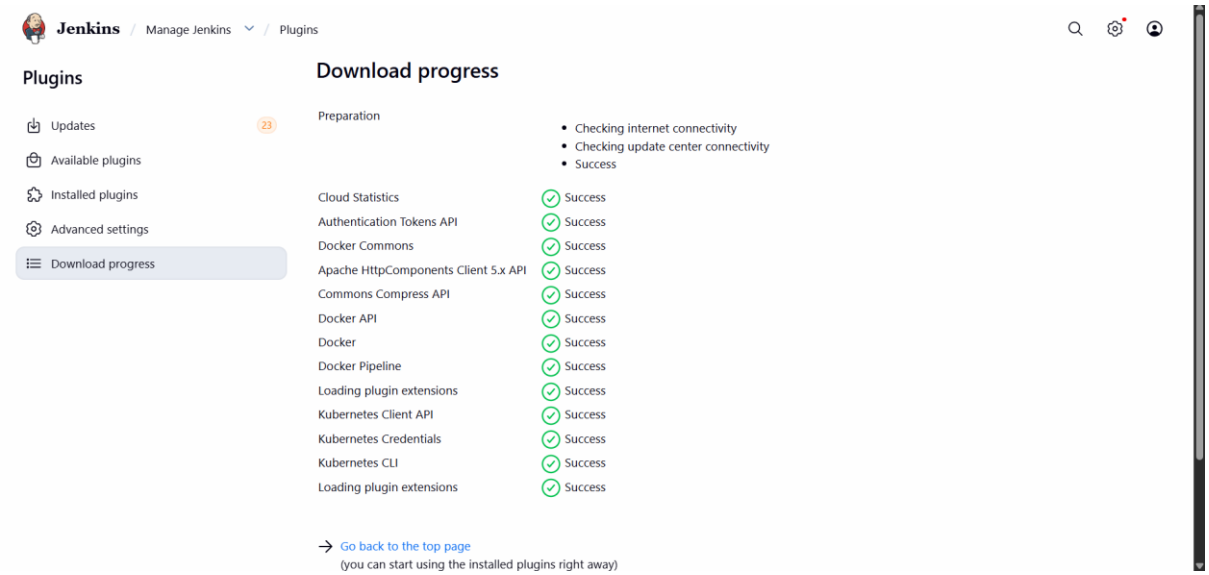
23MIS0070

1. Implement CI and CD using GIT+GITHUB+DOCKER + KUBERNETECS using JENKINS for any application.

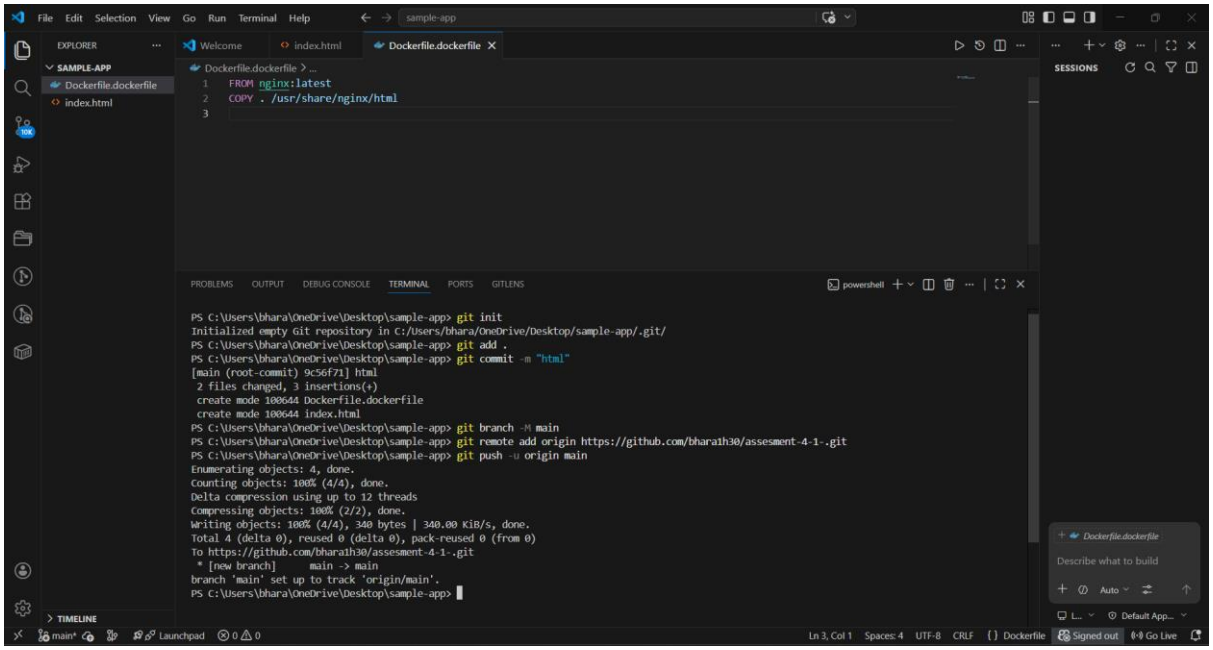
Running kubernetes



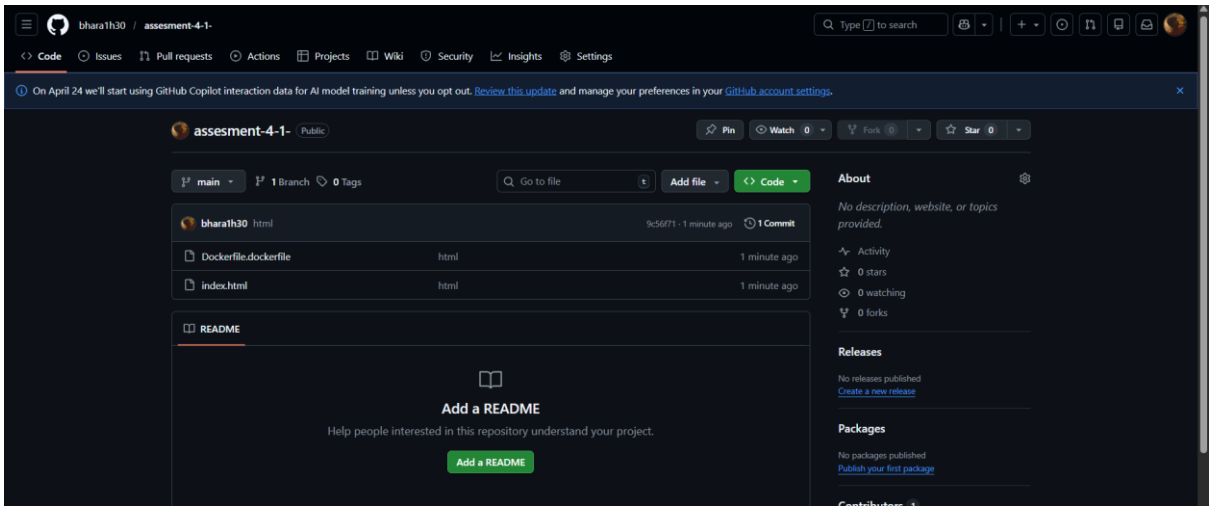
Installing docker, dockerpipeline , Kubernetes



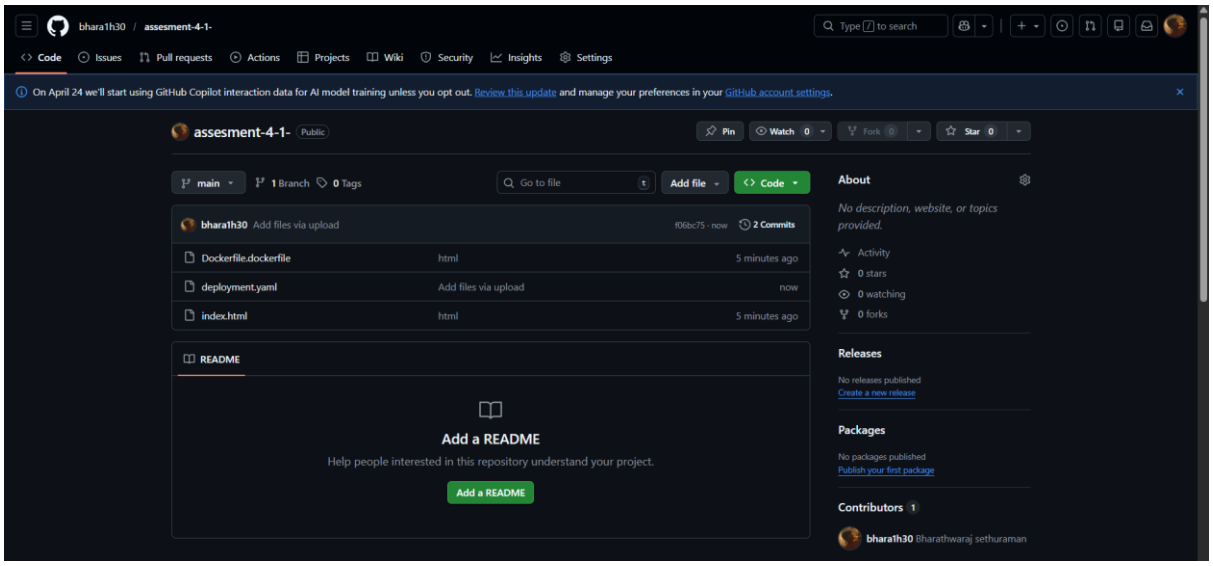
Pushing file:



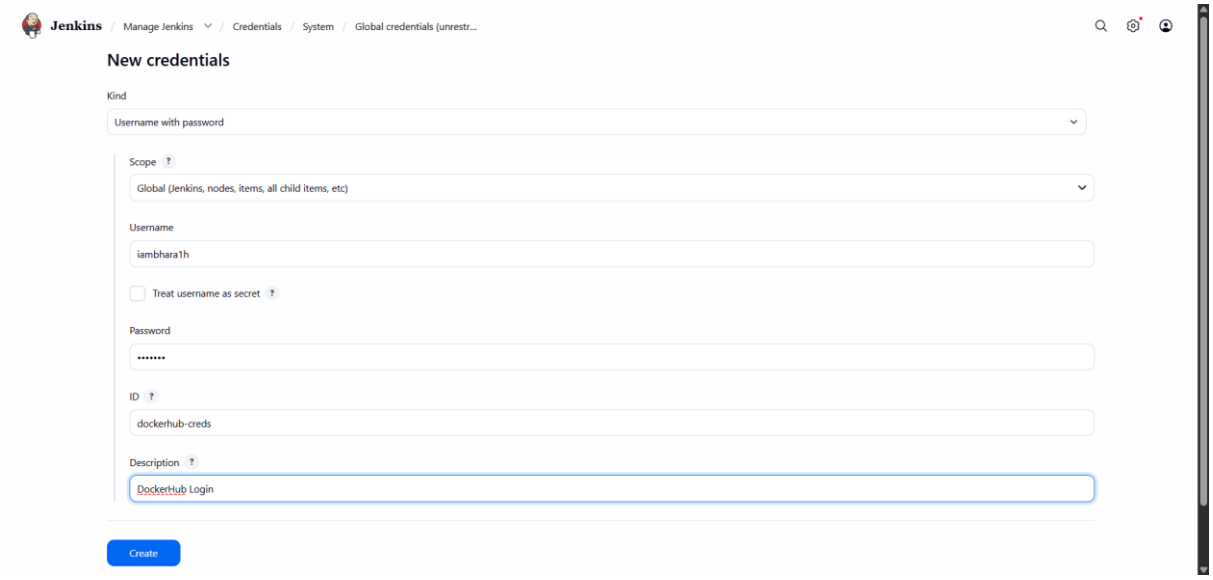
In github:

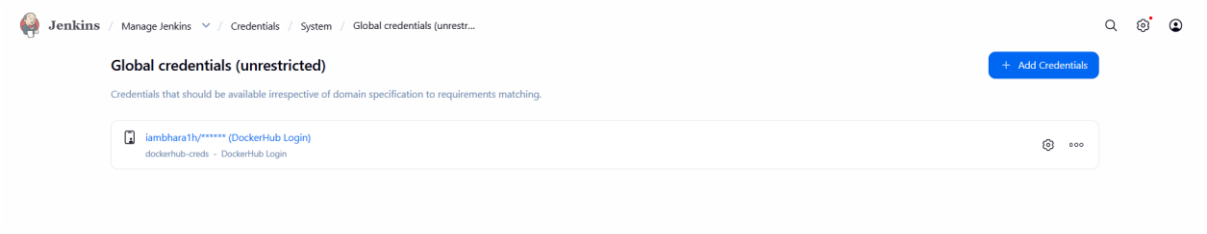


Adding deployment file

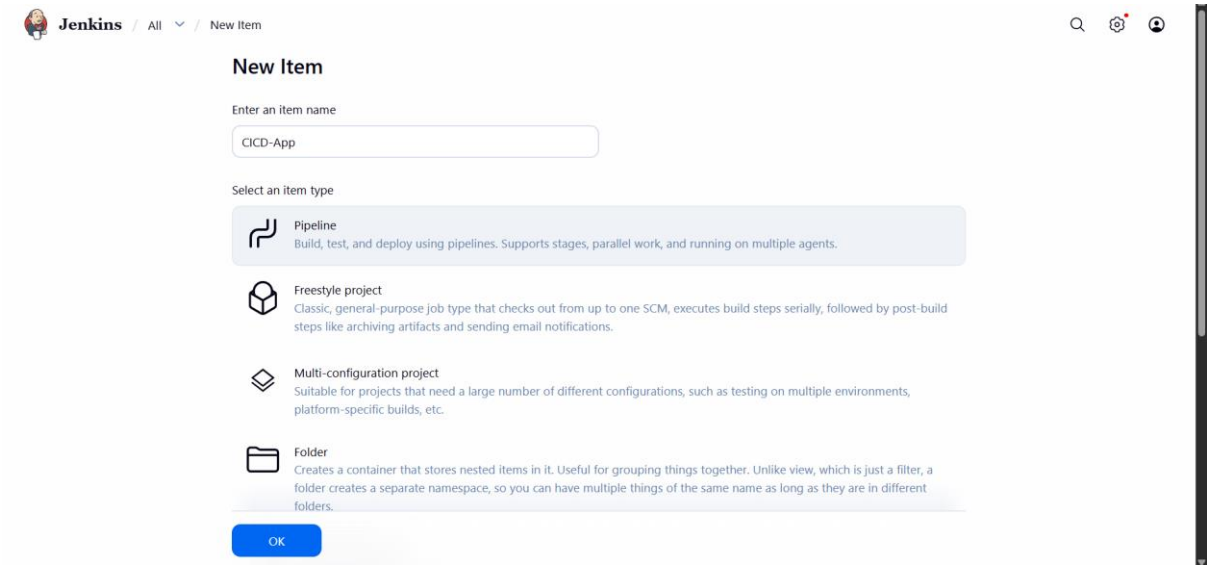


Create credentials

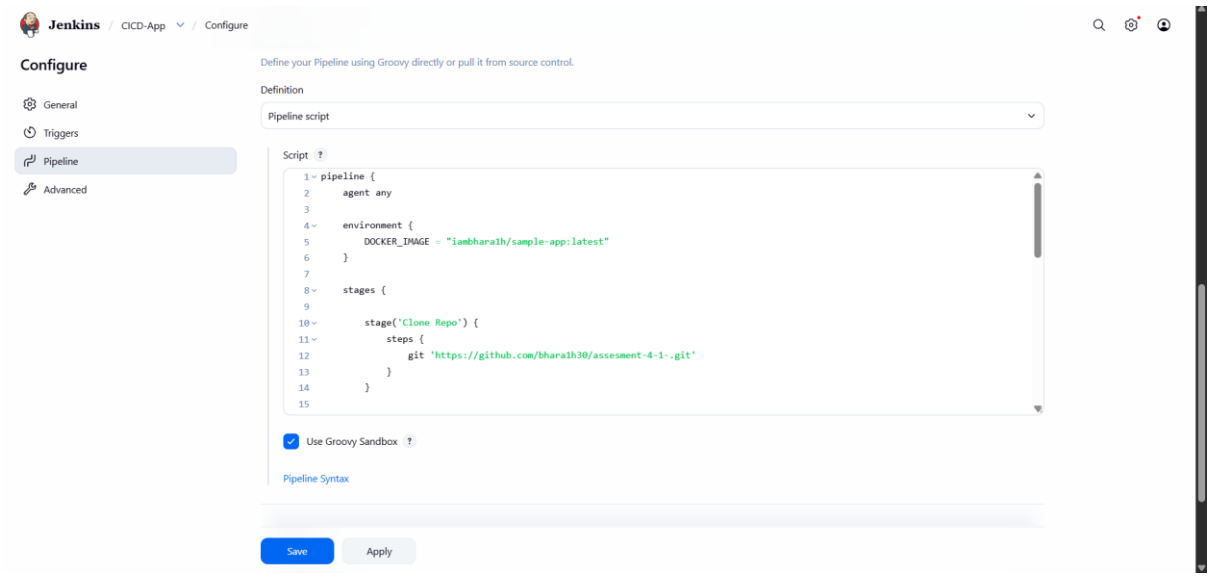




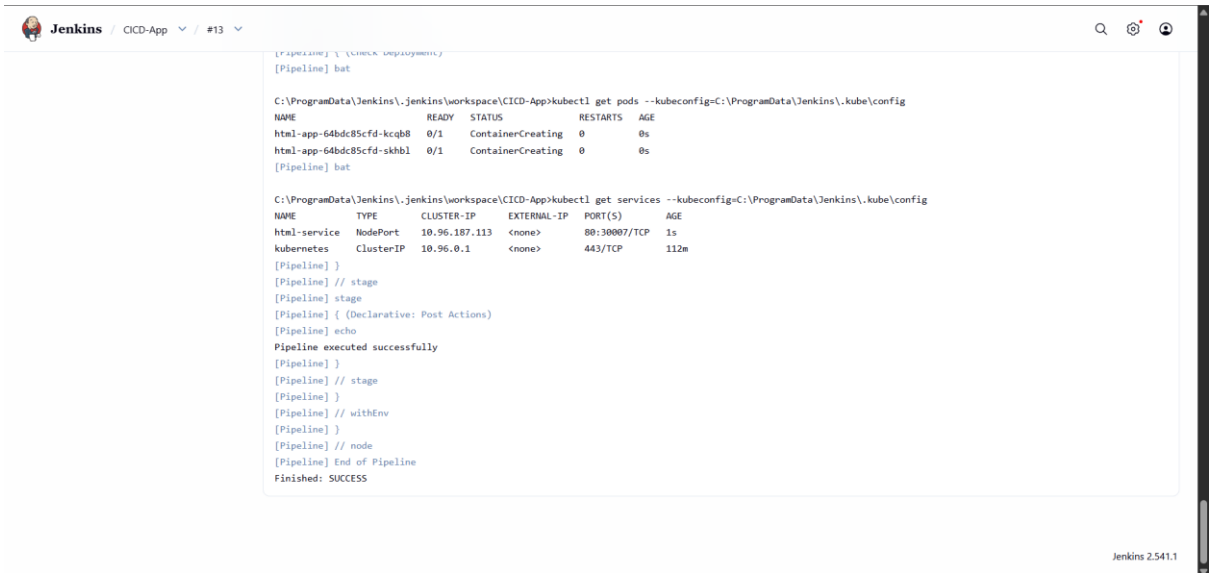
Create pipline in jenkins



Pipe line code



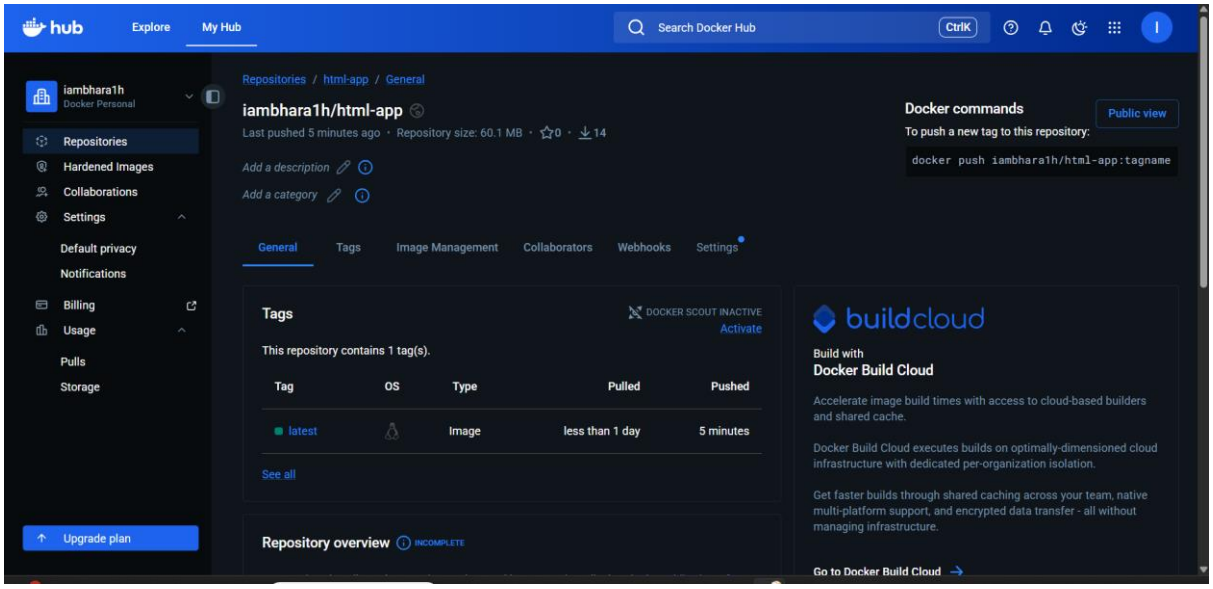
Output in Jenkins



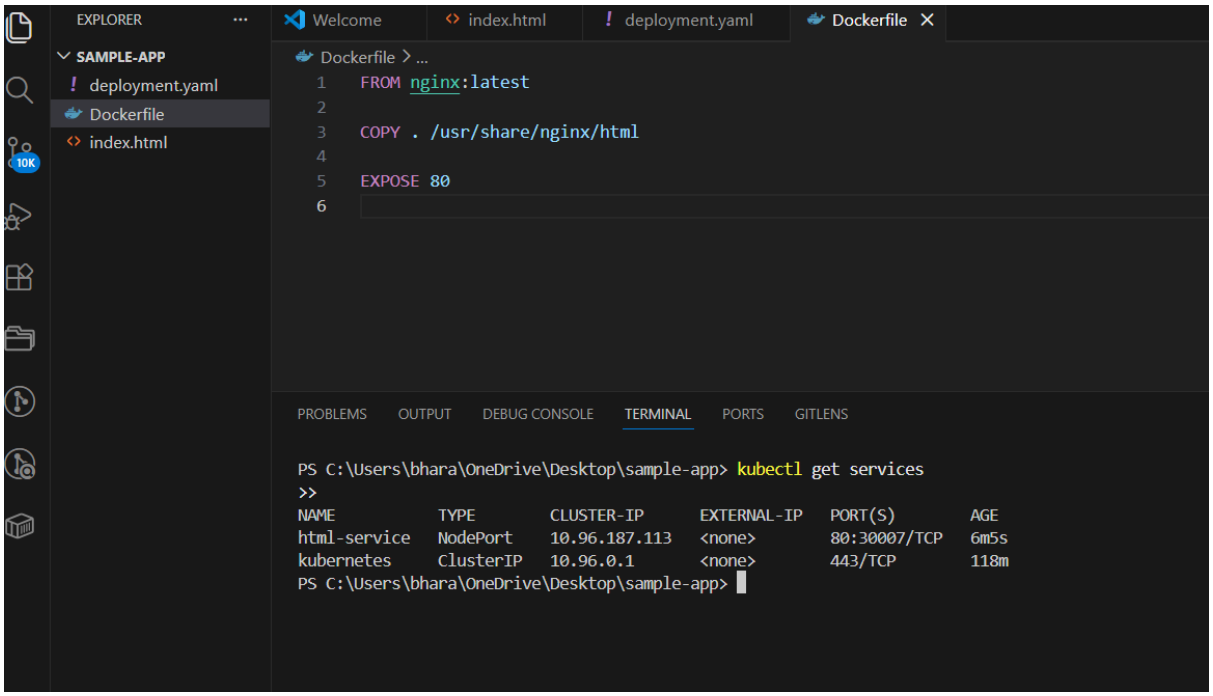
Code:

```
pipeline {
  agent any

  environment {
    DOCKER_IMAGE = "iambhara1h/html-app:latest"
    KUBECONFIG = "C:\\ProgramData\\Jenkins\\.kube\\config"
  }

  stages {

    stage('Clone Code') {
      steps {
        git branch: 'main',
          url: 'https://github.com/bhara1h30/assesment4.git'
      }
    }

    stage('Build Docker Image') {
      steps {
        bat 'docker build -t %DOCKER_IMAGE% .'
      }
    }

    stage('Push Image') {
      steps {
        withCredentials([usernamePassword(
          credentialsId: 'dockerhub-creds',
          usernameVariable: 'USER',
          passwordVariable: 'PASS'
        )]) {

          bat """
          docker login -u %USER% -p %PASS%
          docker push %DOCKER_IMAGE%
```

```
            """
            }
        }
    }

    stage('Deploy to Kubernetes') {
        steps {
            bat 'kubectl apply -f deployment.yaml --kubeconfig=%KUBECONFIG% --validate=false'
        }
    }

    stage('Check Deployment') {
        steps {
            bat 'kubectl get pods --kubeconfig=%KUBECONFIG%'
            bat 'kubectl get services --kubeconfig=%KUBECONFIG%'
        }
    }
  }

  post {
    success {
        echo 'Pipeline executed successfully'
    }

    failure {
        echo 'Pipeline failed'
    }
  }
}
```

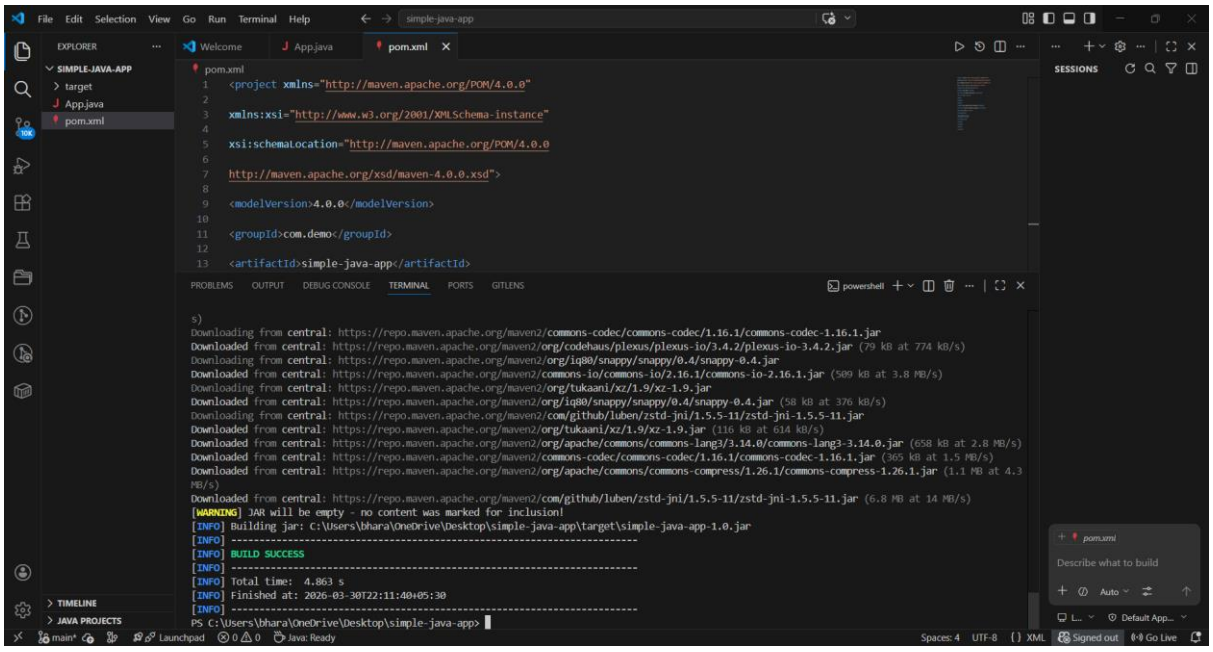


Run this command to verify in command prompt

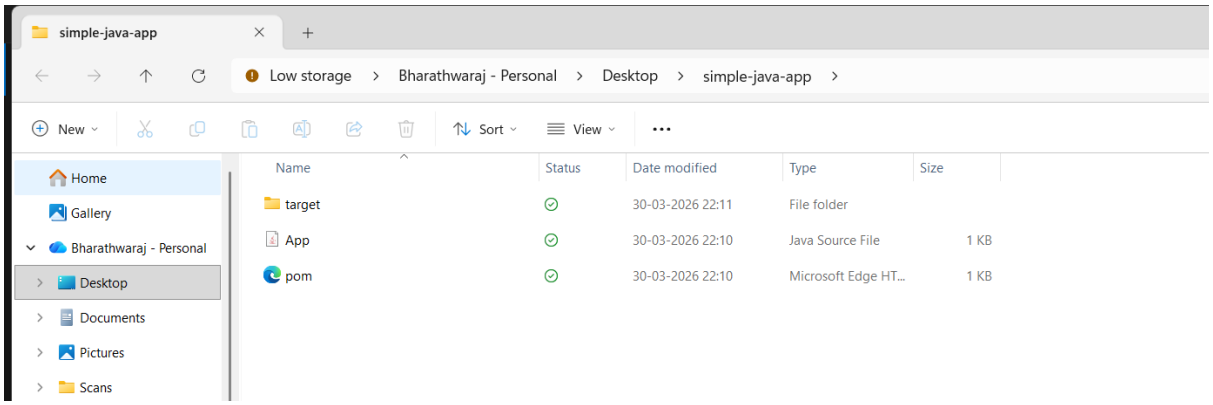


2. Implement CI and CD using GIT+GITHUB+MAVEN+ DOCKER + KUBERNETECS using JENKINS for JAVA application

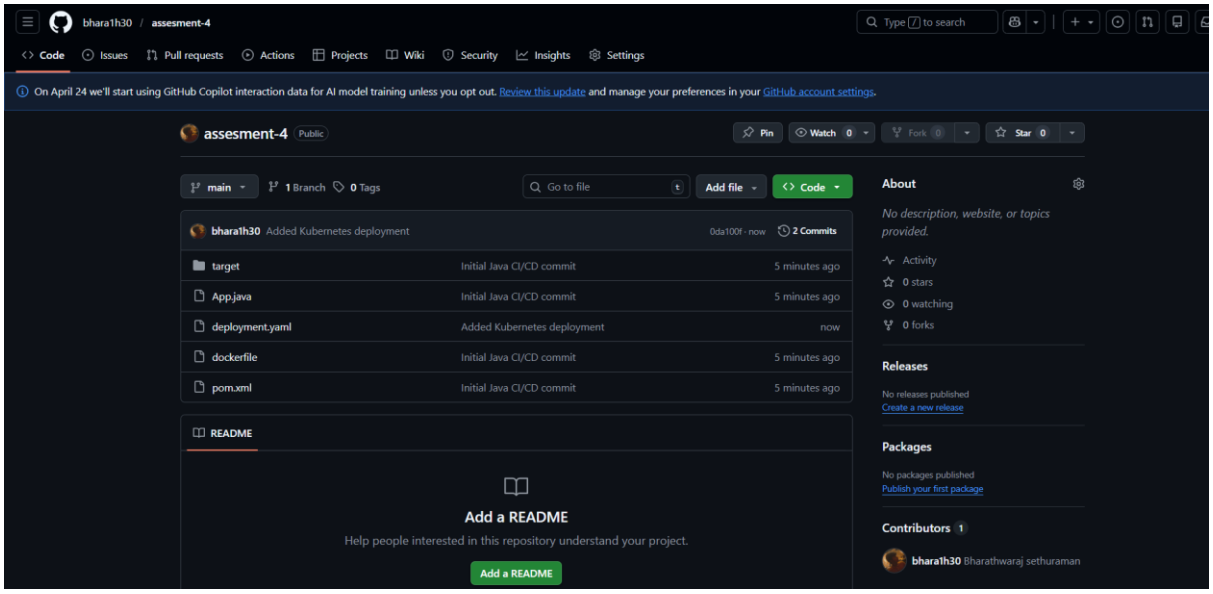
Maven



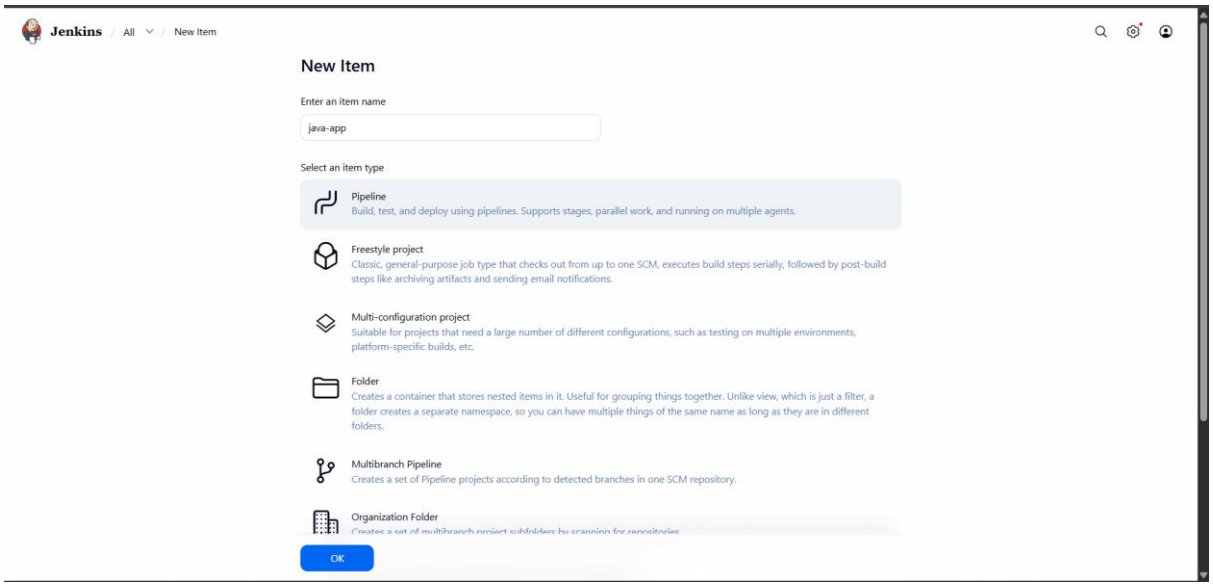
Target creates using mvn clean



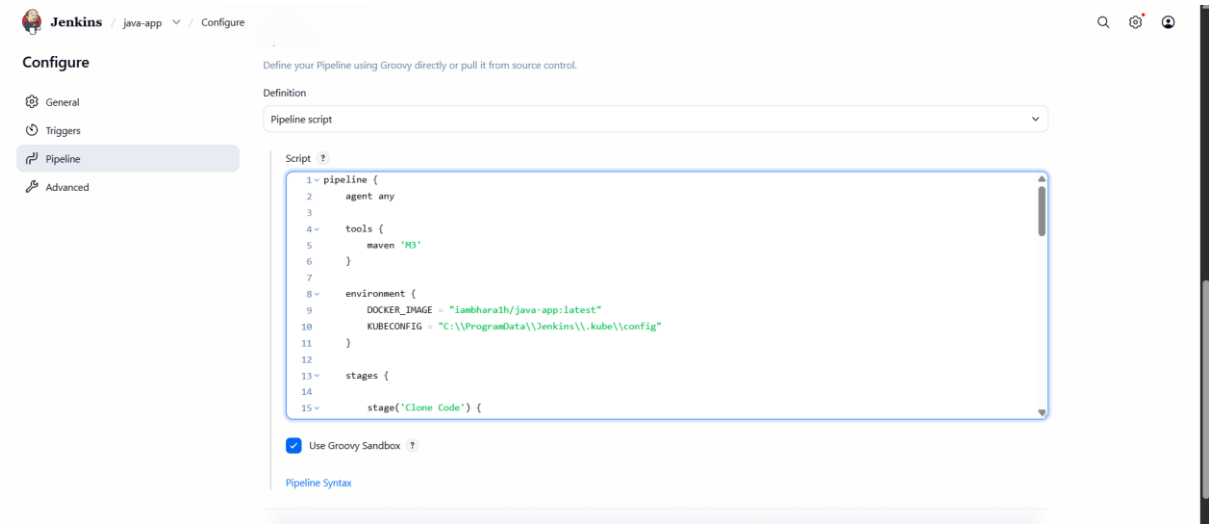
Files pushed in git:



Jenkins for java-app



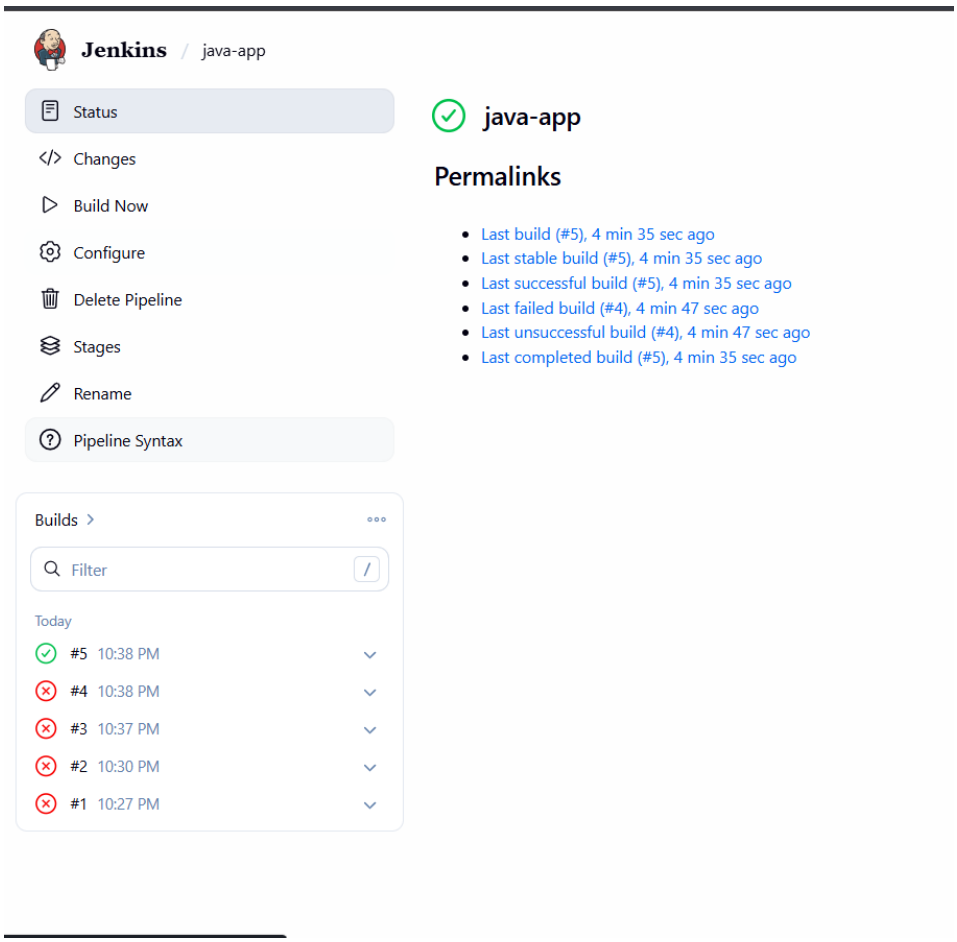
Jenkins code:



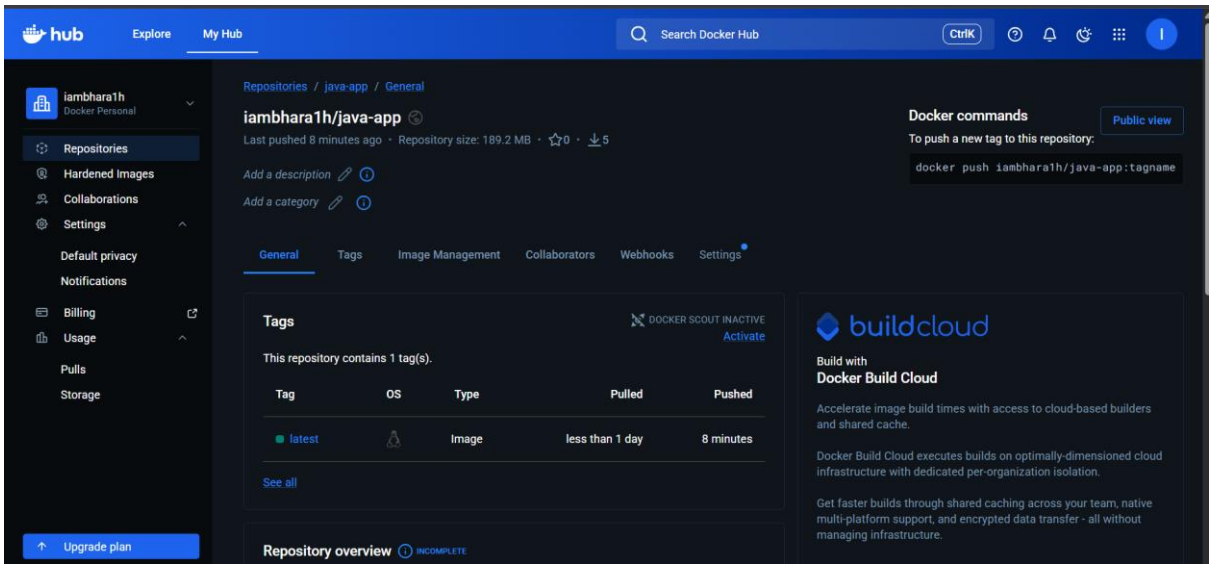

```
pipeline {
    agent any

    tools {
        maven 'M3'
    }

    environment {
        DOCKER_IMAGE = "iambhara1h/java-app:latest"
        KUBECONFIG = "C:\\ProgramData\\Jenkins\\.kube\\config"
    }

    stages {

        stage('Clone Code') {
            steps {
                git branch: 'main',
```

```
            url: 'https://github.com/bhara1h30/assesment-4.git'
        }
    }

    stage('Maven Build') {
        steps {
            bat 'mvn clean package'
        }
    }

    stage('Build Docker Image') {
        steps {
            bat 'docker build -t %DOCKER_IMAGE% .'
        }
    }

    stage('Push Image') {
        steps {
            withCredentials([usernamePassword(
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'USER',
                passwordVariable: 'PASS'
            )]) {

                bat '''
                echo %PASS% | docker login -u %USER% --password-stdin
                docker push %DOCKER_IMAGE%
                '''
            }
        }
    }

    stage('Deploy to Kubernetes') {
        steps {
            bat 'kubectl apply -f deployment.yaml --kubeconfig=%KUBECONFIG% --validate=false'
        }
    }
  }

  post {
    success {
        echo 'Java CI/CD Pipeline Successful'
    }
    failure {
        echo 'Pipeline Failed'
    }
  }
}
```

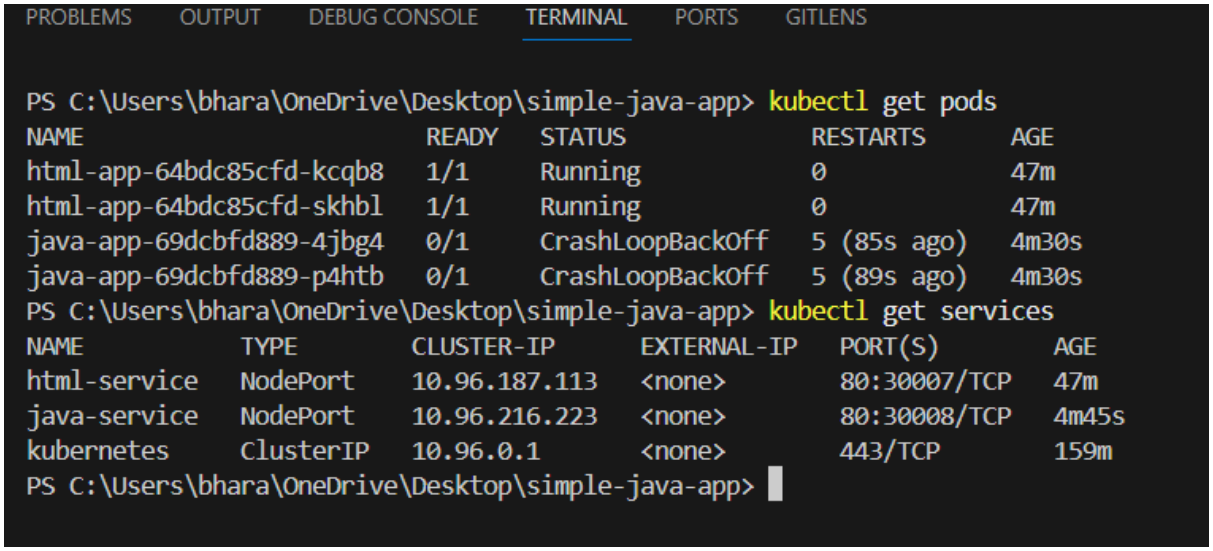
Jenkins output



In docker



Run this command to verify in command prompt



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS

PS C:\Users\bhara\OneDrive\Desktop\simple-java-app> kubectl get pods
NAME                        READY   STATUS             RESTARTS      AGE
html-app-64bdc85cfd-kcqb8   1/1     Running            0             47m
html-app-64bdc85cfd-skhbl   1/1     Running            0             47m
java-app-69dcbfd889-4jbg4   0/1     CrashLoopBackOff   5 (85s ago)   4m30s
java-app-69dcbfd889-p4htb   0/1     CrashLoopBackOff   5 (89s ago)   4m30s
PS C:\Users\bhara\OneDrive\Desktop\simple-java-app> kubectl get services
NAME           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
html-service   NodePort    10.96.187.113   <none>        80:30007/TCP   47m
java-service   NodePort    10.96.216.223   <none>        80:30008/TCP   4m45s
kubernetes     ClusterIP   10.96.0.1       <none>        443/TCP        159m
PS C:\Users\bhara\OneDrive\Desktop\simple-java-app>
```